UNIT 3 MATERIAL REQUIREMENTS PLANNING (MRP)

Structure

3.1 INTRODUCTION

In a large-scale industry, materials management is so important that non-availability of required material leads to several problems, such as stoppage of production, loss of reputation due to not meeting the demand, loss of worker efficiency, and so forth. To avoid this and for smooth flow of production, a good coordinated materials management system is required. Material Requirements Planning (MRP) is such a tool that integrates all the connected people of materials. Material Requirements Planning is a management tool to ensure that materials and components/parts are available at the right time so that the finished products can be completed according to master production schedules.

According to American Production and Inventory Control Society (APICS):

"MRP constitutes a set of techniques that use bill of materials (BOM), inventory data, and master production schedule (MPS) to calculate the requirement of materials for making a good line of balance (LOB)."

MRP is an internal production process designed to ensure that materials (including raw materials, components, sub-assemblies, supplies for the production operations, maintenance, and repair purposes) and their parts are available as and when required.

Objectives: After reading this unit, you should be able to:

- understand jargon of MRP
- explain the inputs and outputs of MRP
- describe the MRP procedure
- state the objectives, limitations and benefits of MRP

3.2 HISTORY AND OBJECTIVES OF MRP

Brief History

MRP initially devised by the General Electric and Rolls Royce, makers of aero engine, during 1950s but not commercialized by them. Joseph Orlicky later on developed and commercialized MRP concept in the year 1964, as a response to the Toyota Manufacturing Program. Black & Decker was the first company to use the concept of MRP in 1964 with Dick Alban as project leader. Almost 700 companies have been implemented MRP by 1975. The number has been increased to about 8000 by the year 1981. Oliver Wight, in 1983, took MRP to next level and developed the philosophy of manufacturing resource planning (MRP II), also known as material planning. By 1989, about one third of the software industry was MRP II software sold to American industry. The timeline of MRP development is shown in figure 3.1.

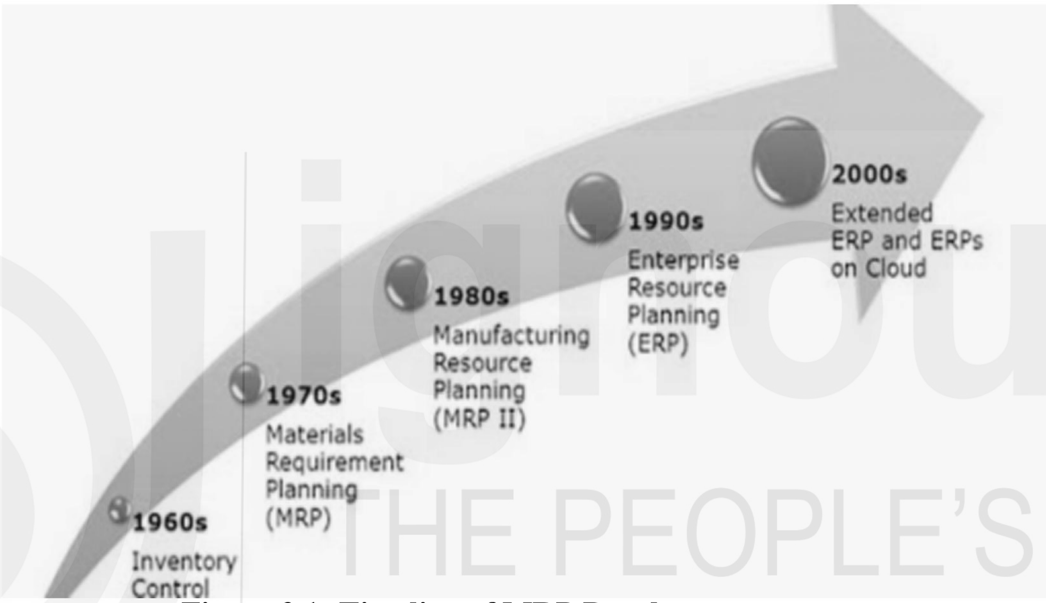

Figure 3.1: Timeline of MRP Development

Objectives of MRP

MRP is designed to enhance the inventory efficiency of a business/organization by estimating quantities of raw material and scheduling timely deliveries. Its primary purpose is to keep adequate inventory levels, so that to required raw materials are available as they are needed. There are three basic objectives of material requirement planning (MRP) as stated below:

1. To ensure required materials are available for manufacturing and products/goods are available for delivery to customers
2. To maintain the lowest possible material and product levels in warehouse
3. To plan production activities, delivery schedules and purchasing activities

3.3 THE JARGON OF MRP

There are few terms/definition that will come up in MRP repeatedly. These terms are as follows:

1. **Item:** An item is the name or code number used for the event which is being scheduling.

2. **Lot Size:** This is the quantity of units you order during manufacturing
3. **Lead Time:** This is the time you need to assemble or manufacture an item from beginning to end.
4. **Ordering lead time**: It is the time taken from starting the purchase to receiving the purchase.
5. **Manufacturing lead time**: It is the time taken for the company to completely manufacture a product from start to end.
6. **Cumulative Lead Time:** This is the greatest amount of time that it takes to develop the product.
7. **Master Production Schedule:** This is the schedule of finished products that drives the MRP process. The quantities in MPS represent what you need to produce to meet the forecast.
8. **Scheduled Receipts (SR):** These are the open orders for products that the company currently possesses but has not yet fulfilled.
9. **Product Structure Tree:** This is a visual representation of the bill of materials, showing how many of each part and how many sub-parts required to produce the product.
10. **Net-Change Systems:** These are systems which identify only the changes between the new and old plan.
11. **Lumpiness:** This is when product/material that is low or at zero suddenly spikes. Examples of lumpy or uneven demand include the need for service parts. You only need service parts when an appliance breaks, so forecasting the need for the parts may be difficult, as the demand is not continuous.
12. **Time Fence:** Time fences are boundaries between different MRP planning periods. They offer the opportunity for programming changes, such as rules and restrictions.
13. **Low-Level Code:** This is the lowest level code of an item in the bill of materials and indicates the sequence in which you run items through an MRP.
14. **Past Due:** This is the time during which you consider orders behind schedule.
15. **Gross Requirements (GR):** This is the total demand for an item during a specific time period.
16. **Projected on Hand (POH):** This is the amount of inventory that is estimated to be available after meeting the gross requirements.

$$\textit{Current POH} = \textit{Previous POH} + SR + POR - GR$$

17. **Net Requirements (NR):** This is the actual, required quantity to be produced in a particular time period.
18. **Planned Order Receipts (POR):** The quantity of orders during a time period that is expected to be received. This planning for orders keeps the inventory from going below the threshold necessary.
19. **Planned Order Releases (PORL):** This is the amount a company plan to order per time period. This is POR offset by the lead time.

3.4 FUNCTIONS OF MRP

Companies following a systematic framework for planning material may keep their production department, inventory control and purchasing department in order and ensures smooth flow of material through facility. MRP system may not run a production facility by its own, but it allows the company to maintain a steady flow of material through supply chain. Some of the functions of an MRP system include:

i. Inventory Management

ii. Cost Reduction

iii. Production Optimization

i. **Inventory Management**: Primary function of MRP is to look that the raw materials are available when they are required. This in turn helps in ensuring that company does not have too little or too much material in store. Inventory management is needed as maintaining too much material will incur storage costs and maintaining too little inventory leads to delays and shortages.
ii. **Cost Reduction**: By using MRP system effectively, there will be a significant reduction in cost for manufacturing facilities. First, by minimizing the time that the managers spend on manually calculating the quantities and time for each material. Second, via inventory management, as MRP will assure that the money is not lost by storing unnecessary material.
iii. **Production Optimization:** As MRP is intended to manage materials, it acts as a prominent tool to improve production process. When the material/parts/items flow through the production facility, companies can save time and decrease costs. Men and machines work with consistency at faster rate.

3.5 NATURE OF DEMANDS

There are different sources of demand for a product and its component items. Few item requirements are determined by the needs of other items while others are specified by customers. Item requirements can be classified as dependent and independent demands. Independent demand is a desire for final end products, such as laptops or motor bikes, whereas dependent demand is the demand for parts, components, or incomplete assemblies such as screen guards or tires. Companies estimate quantities for the dependent demand by determining quantities for the independent demand. For instance, if a company can forecast independent demand for laptops that are expected to sell, then they can forecast the quantities of dependent demand materials such as screen guards. The relationship between the materials and the finished product are shown on a bill of materials (BOM) and are calculated with MRP.

SAQ-3.1

a) State the objectives of MRP
b) Differentiate between dependent and independent demand.
c) What are the functions of MRP? Explain
d) Define the following terms
    i. Gross Requirements
    ii. Net Requirements
    iii. Planned Order Receipts
e) What do you understand by "Lumpiness" and "Time Fence"?

ACTIVITY-3.1

Visit any paper tea cup manufacturing unit located in your locality. Study the manufacturing process and prepare Master Production schedule for tea cup manufacturing

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3.6 STEPS IN MATERIAL REQUIREMENT PLANNING

MRP is a well-designed framework of processes and calculations. It helps the companies to transform its operational procedures. People in the organization contribute to the MRP process that includes production, sales, purchasing, stockroom, receiving and shipping. MRP includes three basic steps. They are

i. Identification of quantity requirements
ii. Performing MRP calculations
iii. Completing the orders.

i. Identification of quantity requirements

The quantity is on hand, planned for production, committed to current orders and forecasted are to be determined initially.

ii. Performing MRP Calculations

Suggestions are created for materials that company considers critical, expedited, and delayed.

iii. Complete the Orders

List and describe the materials for the purchase orders, manufacturing orders, and other reporting requirements.

3.7 MRP INPUTS AND OUTPUTS

The calculations that MRP performs are based on the data inputs. Inputs for MRP are master schedules, bill of materials, and inventory status records, whereas outputs for MRP are purchase orders, material plan, work orders and reports. These are diagrammatically represented in Figure 3.2 and briefly discussed in the following paragraphs.

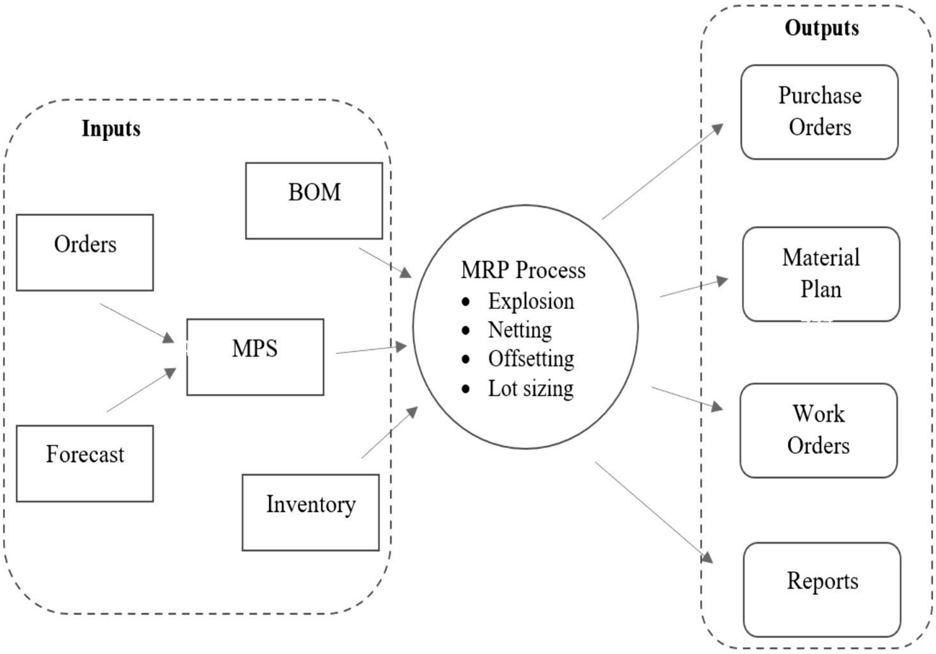

Figure 3.2: Inputs and outputs of MRP

Inputs of MRP

After going through the figure 3.2, one can easily notice that the following are the inputs required for MRP:

i. Customer orders
ii. Forecast demand
iii. Master production schedule (MPS)
iv. Bill of material (BOM)
v. Inventory Records

i. **Customer Orders:** Customer orders refer to the specific information that the organization may receive from customers and includes one-offs and regular ordering patterns.

ii. **Forecast Demand:** This predicts how much probable demand there will be for a product or service. Basically, forecast demand is based on historic data and present trend.

iii. **Master Production Schedule (MPS):** Both forecast demand and customer orders feed into the master production schedule. The MPS is a plan that a company develops for production, staffing, or inventory. It is a comprehensive schedule of what is to be produced in a given period based on the accurate estimate of demand. It is a list of end products to be produced in a given period. It also includes production costs, inventory costs, inventory information, supply, lot size, lead time, and development capacity. An exemplary MPS used in construction and shop floor production shops is shown in Tables 3.1 and 3.2, respectively, in two different models.

Table 3.1: Exemplary master production schedule for construction of a building

Month	1	2	3	4	5	6	7	8	9	10	11	12
Foundation												
Pillar construction												
Slab												
Construction of walls												
Flooring												

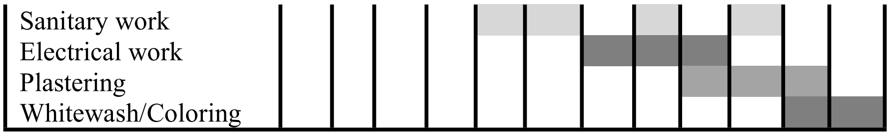

Table 3.2: Production schedule in a XYZ shop floor

Month	April				May		
Week Number	**1**	**2**	**3**	**4**	**1**	**2**	**3**
Product 1		600	300		1200		200
Product 2	800	200				400	100
Product 3			500	800		400	500

iv. **Bill of Materials (BOM):** It is one of the three primary inputs of MRP and is a list of parts or components required for producing the final product, showing all the details of its required quantity at every level of its parts assembly/subassembly. The visual depiction of the requirements in a bill of materials where all components are listed by levels is called product structure tree (PST). The final product is designated generally by “0 level” while the raw material stage at the 10th level in BOM is as shown in the exemplary Figure 3.3

For a better understanding of PST and BOM, consider an example of a clipboard. General assembly clipboard components, their product structure tree, along with the levels of bill of materials are shown in Figure 3.4. Finally, the bill of materials for the clipboard is prepared as shown in Table 3.3.

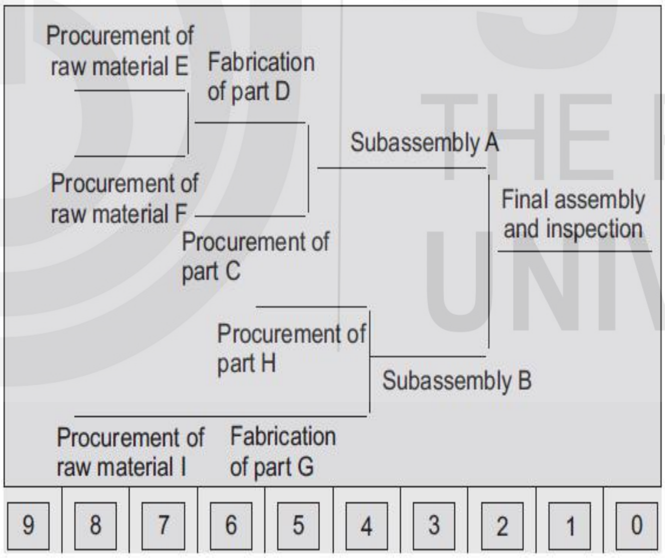

Figure 3.3: Exemplary product structure tree showing levels of BOM

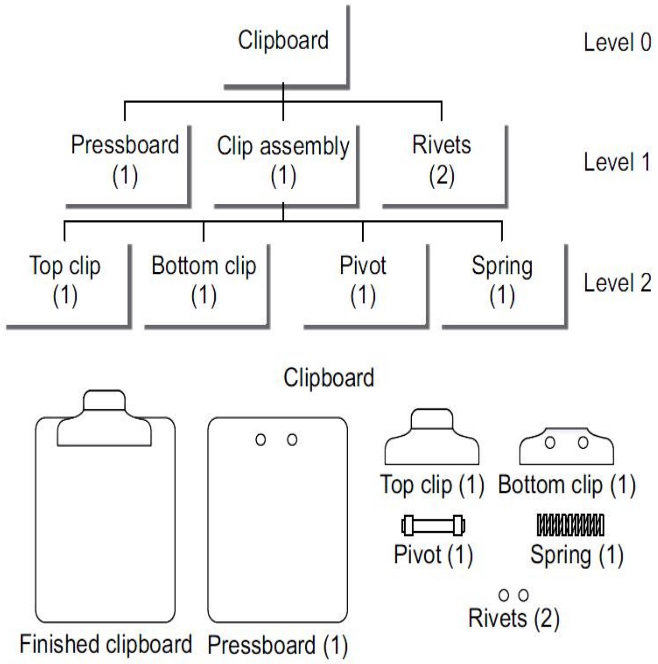

Figure 3.4: PST with levels of BOM for the components/ assemblies of clipboard

Table 3.3 Bill of materials for clipboard

Part Name	Assembly Level	Quantity
Clipboard	0	1
Pressboard	1	1
Clip Assembly	1	1
Rivets	1	2
Top clip	2	1
Bottom clip	2	1
Pivot	2	1
Spring	2	1

v. **Inventory Records:** These are the raw materials and the completed products that you either have on hand or have already ordered. An exemplary inventory record of welding electrodes of a hypothetical company is given in Table 3.4 for reference.

Table 3.4: Sample inventory record of welding electrodes

Days	1	2	3	4	5	6	7
Gross Requirement	120	-	20	-	150	50	-
Schedule receipt/in transit	50	20	-	30	70	50	-
On hand	70	50	50	60	50	-	-
Net requirement	70	-	10	40	120	50	-
Planned order release	80	-	10	50	120	50	-

Integrity of MRP Input Data

Data integrity refers timeliness, completeness, and accuracy. Input data for MPR should be provided by the concerned people and/or machine in time and accurately. If the data is entered into the system is incorrect, MRP may lead to false information. MRP should provide the organization and manager with credible data, but the presence of errors may destroy the credibility and results to ridiculous plan. Further, attitude, Discipline, and training are the keys entities to data integrity. Education of people in the company is the most important factor.

Auditing of data processing must be done at regular intervals to keep the data valid. Top Management should look after the discipline, training and motivation of employees who handles data. All the employees in the company who are handling data should accept responsibility for quality of data handled. The purpose of data integrity is to identify and remove the causes of errors. Organizations using MRP systems should incorporate self-checking, self-correcting and auditing features into the systems. Automatic data integrity checks of input data include the following types of tests

a) reasonableness test (e.g. abnormal quantity or unit-of-measure),
b) existence test (e.g. part number, transaction code),
c) internal detection (e.g. negative inventory balances),
d) diagnostic test (e.g. prior transactions required), and
e) purging residences of undetected errors (e.g. closing out old shop or purchase orders)

Outputs of MRP

After MRP receives the input, it generates the output. Outputs of MRP are categorized into four main parts. These include:

i. Purchase Orders:
ii. Material Plan
iii. Work Orders
iv. Reports

- **Purchase Orders:** It is the purchasing schedule that includes the order a company gives to the supplier to supply the material. It also include the details like quantities, start and finish dates.
- **Material Plan:** Master plan includes the details like material, assembly parts and components that are required to finish the end product with quantities and schedule
- **Work Orders:** This includes the work concerned with producing the end product including details of materials required, start and finish dates of the end product, and department wise responsibilities.
- **Reports:** MRP generates two types of reports i.e, primary and secondary reports. Purchase order, material plan and work orders are included in primary reports. Secondary reports details performance control, data errors, deviations and future inventory and contract predictors

SAQ-3.2

a) What are the steps in MPR process? Explain
b) Describe the inputs and outputs for MRP.
c) What is mater production schedule? What is its role in MRP?
d) Explain Bill of Materials (BOM) with an example.
e) Why integrity of MRP input data is required?

ACTIVITY-3.2

Prepare the bill of materials (BOM) for spectacles/sunglasses/goggles. Also draw the product tree structure and represent the level codes.

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3.8 MRP PROCESS

The core part of MRP system is the mechanism that transforms input data into output. This transformation of inputs to outputs in performed systematically, following a sequential steps namely explosion, netting, offsetting and lot sizing. The flow of these steps is shown in the figure no 5. In the explosion, the disassembly of parts of end products is simulated. In netting process, gross requirements are adjusted to account for on-hand inventory or quantity on order. This adjustment is done at every level of the Bill of Materials and for each time bucket. In offsetting, the timing of order released is calculated. In order to meet net requirements, an order release is offset by the production lead time or supplier delivery time. Finally, the lot sizing is the step in which the batch size to be purchased is determined. The detailed explanation of each step in explained through figure 3.5 below.

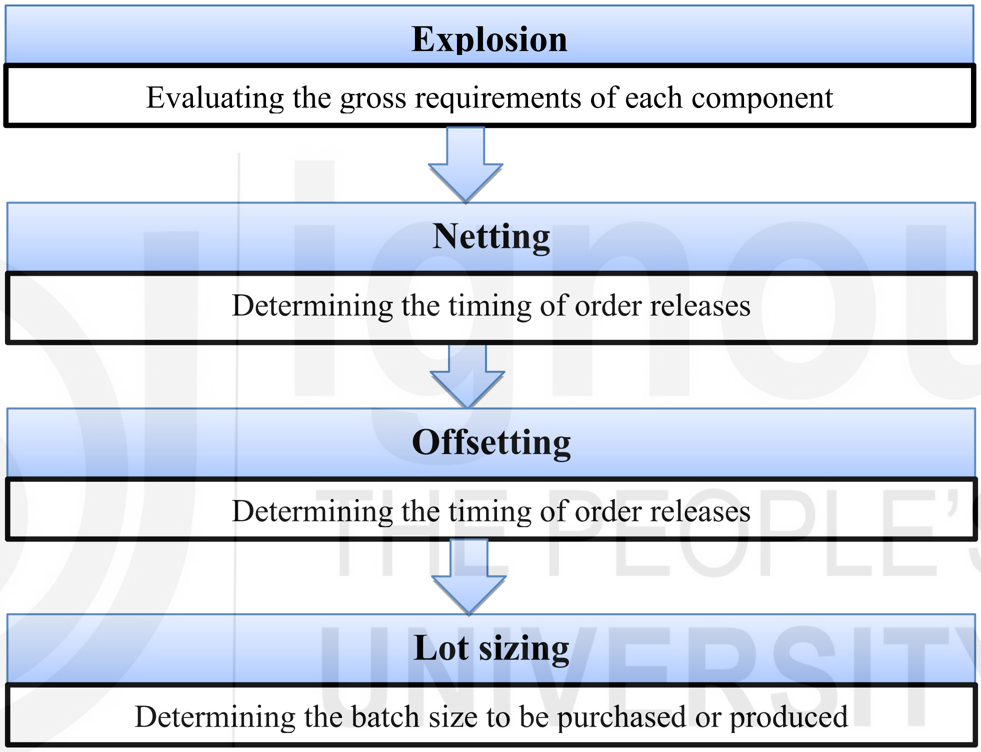

Figure 3.5: MRP Process

Exploding

This step uses the Bill of Materials (BOM). In this, components and its quantities required for each item is listed. Generally BOM's are characterized by the number of levels it involves, followed by the structure of assemblies and sub assemblies. The first level is represented by the MPS and is 'exploded' down to final assembly. Thus a given number of finished products are exploded to see how many items are required at the final assembly stage.

Netting

The next step is 'netting', in which any stock on hand is subtracted from the gross requirement determined through explosion, giving the quantity of each item needed to manufacture the required finished products.

Offsetting

In offsetting, the timing of order released is calculated. In order to meet net requirements, an order release is offset by the production lead time or supplier delivery time.

Lot sizing

Lot sizing is the step in which the batch size to be purchased or produced is determined.

3.9 MRP PROCEDURE

Master Production Schedule procedure reinforces the independent demands of forecasts and customer orders in order to determine the requirements of the finished products in each time bucket. Next after netting the on-hand and on-order inventory, and offsetting the lead-time, the production schedule of the end products is determined. Further, the available-to-promise (ATP) is also determined. Now, MPS is fed into the MRP procedure to determine the requirements of the Lower Level Components (LLC) and raw materials. The gross requirements (GR) of parts are estimated by determining the planned order releases (POR) of the parents through single level BOM explosion. The net requirements are then calculated by subtracting the on-hand inventory and scheduled receipts in each time bucket. After the consideration of lot-size, the net requirements are converted into the planned order receipts. Lead-time offsetting shifts the planned order receipts backward and derives the POR which are the MRP result of current item. The MRP procedure move forward to explode the POR to obtain the gross requirements of its components. The MRP procedure is repeated until the POR of all the items are determined. The flow chart of the MRP procedure is described in Figure 3.6.

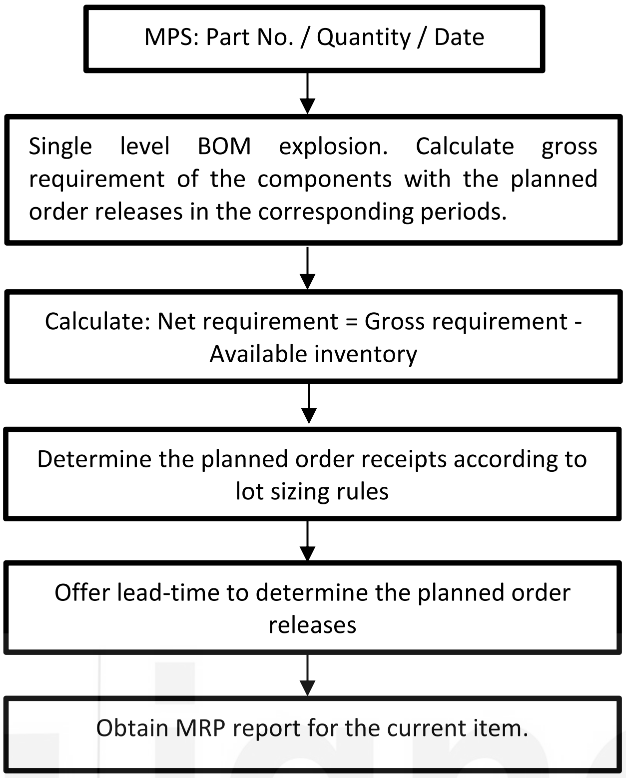

Figure 3.6: Flow chart for MRP procedure

The procedure of MRP can also be studied by categorizing into four different elements, namely triggers, data, tasks and outcomes. These are depicted in the figure no. The brief sequential procedure is also shown in the figure no.

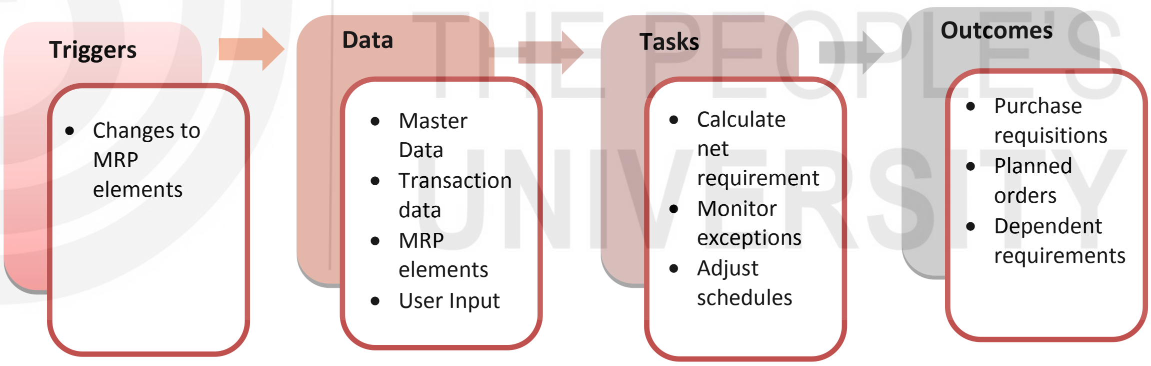

Figure 3.7: Different stages and tasks in MRP

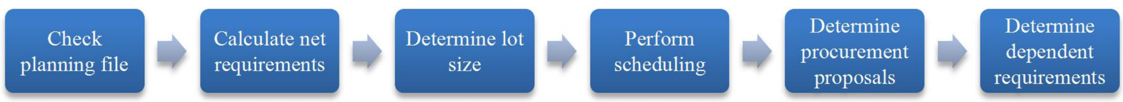

Figure 3.8: Sequential procedure of MRP

3.10 BENEFITS OF MRP

MRP system prioritizes materials based on the need and updates these priorities regularly. Thus, materials are ordered at the right time, and whenever changes occur, the MRP system revises the due dates and schedules with minimum delays. As a result, MRP enables to have right components in the right quantity at the right time. This minimizes unnecessary in-process inventory without any shortages. Further, by planning anticipated material requirements, facilities, machinery, and manpower resources can be effectively utilized. Thus, the benefits due to MRP may be witnessed as follows:

- Reduction in finished goods inventory, in-process inventory, raw material, components and parts, and safety stock.
- Reduction in lead time.
- Reduction in past due orders.
- Advantage due to prioritization.
- Quick response to changes in demand.
- Improved customer service.
- Increased productivity.
- Increased inventory turnover.
- Effective utilization of capacity, machinery, and manpower
- Good control over production department.

3.11 LIMITATIONS OF MRP

1 There should be a valid master production schedule (MPS). Any inaccurate forecasts or sudden changes in demand may result in inaccuracy in MPS.
2 Bill of material and inventory status must be computerized to prevent manipulations.
3 Errors in external lead time (at suppliers) or internal lead time (at manufacturing) cause wrong MRP calculation.
4 Causes overdependence on MRP outputs due to highly computational intensive approach. Any inaccuracy leads to failure of MRP to a large extent.
5 MRP system requires reliable data; otherwise MRP system will become a mess.
6 MRP is more materials management oriented, but it should be a manufacturing or assembly oriented product structure.
7 The success of MRP system requires
    - maintenance of accurate stock records,
    - correct and timely reporting system about the completion of jobs and orders, and
    - integration of all concerned with MRP

3.12 PROBLEMS WITH MRP IMPLEMENTATION

Many sources state that problems associated with MRP systems lie, to some degree, with organizational and behavioral factors. Among the causes cited for MRP system failures, following are few problems associated with MRP implementation.

i. Lack of top management commitment
ii. Insufficient user training and education
iii. Lack of technical expertise
iv. high degree of accuracy for operation requirement

Relaxing MRP Assumptions

1. Inventory is not always the constraining resource.

2. Lead times can vary.
3. Not every transaction requires to be recorded.
4. JIT can be used with MRP.
5. The shop floor may need a more sophisticated scheduling system.

3.13 MATERIAL REQUIREMENTS PLANNING SOFTWARE

MRP software focuses on the period during which a company creates a product, identifies and purchases raw materials, determines resources, and plans the steps for production. The cost of MRP software is a big issue for small businesses, but one need to understand that the investment is a huge money saver in the long run. Some of the features you should consider when choosing MRP software include the following:

- Advanced filtering
- Flexible requirements
- Exception management
- Comprehensive drill-down capability
- Agile order management
- Workflow support
- Supply-chain visibility
- Aggregate requirements
- Multiple quantity display
- Designated time frames
- Forecast consumption
- Software support
- Integration into multiple environments
- Free trials

SAQ – 3.3

a) What are the problems encountered while implementing MRP?

b) List the benefits of MRP

c) State the assumptions of for implementation of MRP.

d) Explain MRP procedure in detail.

e) What are the limitations of MRP?

f) What are the common desirable features of MRP software?

ACTIVITY-2.3

With various (at least three) software providers, conduct an inquiry on the software of MRP and list out what are the features in it. Prepare comparative statement (and determine which software is advisable for an organization of your choice)

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3.14 SUMMARY

Material Requirements Planning is a management tool to ensure that materials and components/parts are available at the right time so that the finished products can be completed according to master production schedules. It is designed to enhance the inventory efficiency of a business/organization by estimating quantities of raw material and scheduling timely deliveries. MRP system may not run a production facility by its own, but it allows the company to maintain a steady flow of material through supply chain. The calculations that MRP performs are based on the data inputs. Inputs for MRP are master schedules, bill of materials, and inventory status records, whereas outputs for MRP are purchase orders, material plan, work orders and reports.MRP transforms inputs to outputs systematically following a sequential steps namely explosion, netting, offsetting and lot sizing.

3.15 KEYWORDS

Bill of materials (BOM): A bill of materials is a list of the raw materials, sub-assemblies, intermediate assemblies, sub-components, parts, and the quantities of each needed to manufacture an end product

Gross Requirements (GR): This is the total demand for an item during a specific time period.

Product Structure Tree: This is a visual representation of the bill of materials, showing how many of each part and how many sub-parts required to produce the product

Master Production Schedule: This is the schedule of finished products that drives the MRP process.

Low-Level Code: This is the lowest level code of an item in the bill of materials and indicates the sequence in which you run items through an MRP.

Netting: It is a process step of MRP where gross requirements are adjusted to account for on-hand inventory or quantity on order.

Projected on Hand (POH): This is the amount of inventory that is estimated to be available after meeting the gross requirement

Net Requirements (NR): This is the actual, required quantity to be produced in a particular time period.

Purchase Orders: It is the purchasing schedule that includes the order a company gives to the supplier to supply the material

3.16 FURTHER READINGS

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